

Stato	Completato
Iniziato	venerdì, 6 giugno 2025, 11:54
Terminato	venerdì, 6 giugno 2025, 13:24
Tempo impiegato	1 ora 29 min.
Valutazione	14,32 su un massimo di 20,00 (71,58%)
Feedback	Thanks for your attempt, all the best!

Domanda 1

Parzialmente corretta

Punteggio ottenuto 1,33 su 2,00

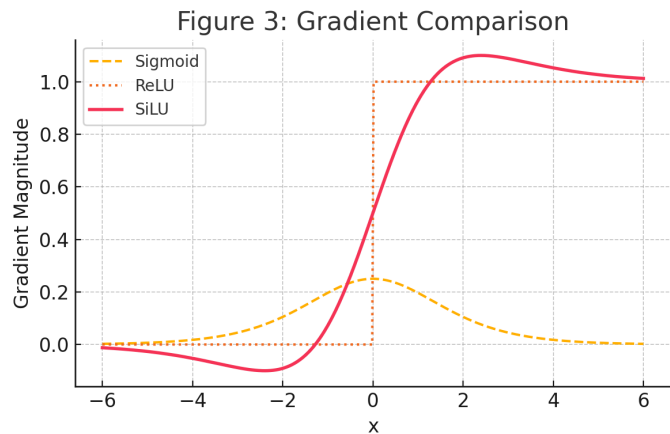
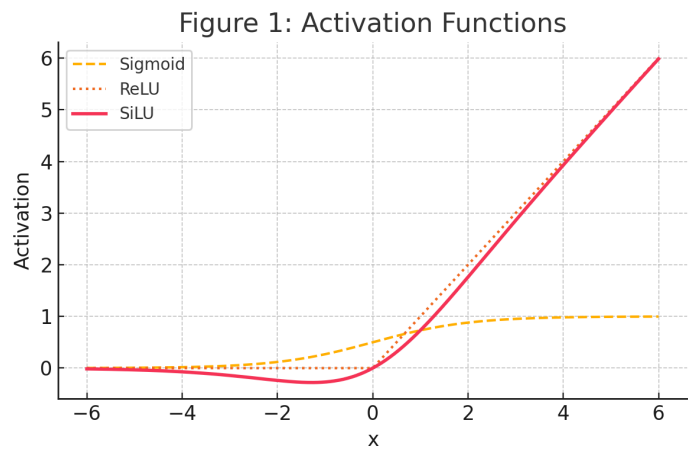
The **SiLU (Sigmoid Linear Unit)** activation function is defined as:

$$\text{SiLU}(x) = x \cdot \sigma(x), \quad \text{where } \sigma(x) = \frac{1}{1 + e^{-x}}$$

Based on this answer the following questions.

True	False		
☒	☐	SiLU is not differentiable at x=0, similar to ReLU.	✓
☒	☐	SiLU is known to suffer from the vanishing gradient problem more severely than sigmoid.	✓
☒	☐	The derivative of SiLU is equal to the sigmoid function: $\frac{d}{dx}\text{SiLU}(x) = \sigma(x)$	✗
☐	☒	For large positive values of x, SiLU behaves approximately like ReLU.	✓
☐	☒	SiLU is a non-monotonic activation function.	✗
☐	☒	SiLU can output negative values for negative inputs.	✓

SiLU has the following shape and gradient



SiLU is not differentiable at x=0, similar to ReLU.: False

SiLU is known to suffer from the vanishing gradient problem more severely than sigmoid.: False

The derivative of SiLU is equal to the sigmoid function: $\frac{d}{dx}\text{SiLU}(x) = \sigma(x)$

: False

For large positive values of x, SiLU behaves approximately like ReLU.: True

SiLU is a non-monotonic activation function.: True

SiLU can output negative values for negative inputs.: True

Domanda 2

Parzialmente corretta

Punteggio ottenuto 0,33 su 2,00

For each of the following, describe if it is a multiclass or multilabel task

Multiclass	Multilabel		
☒ ☐	☐ ☒	Predicting which of the days of the next week my train is likely to arrive late	☐
☐ ☒	☐ ☒	Predicting if a product review is "positive", "neutral", or "negative".	☒
☐ ☒	☒ ☐	A video analysis model where we identify a single action (out of a given list) a person is performing with an object (out of a given list) in a video.	☐
☐ ☒	☒ ☐	Instance segmentation where we indicate, at pixel level, where a person is and identify which of the persons in the image is.	☐
☐ ☒	☒ ☐	Detecting and labeling different kinds of objects in an image with bounding boxes.	☐
☒ ☐	☐ ☒	Predict relevant tags for a programming question on stackoverflow.com.	☐

Predicting which of the days of the next week my train is likely to arrive late: Multilabel

Predicting if a product review is "positive", "neutral", or "negative".: Multiclass

A video analysis model where we identify a single action (out of a given list) a person is performing with an object (out of a given list) in a video.: Multiclass

Instance segmentation where we indicate, at pixel level, where a person is and identify which of the persons in the image is.: Multiclass

Detecting and labeling different kinds of objects in an image with bounding boxes.: Multiclass

Predict relevant tags for a programming question on stackoverflow.com.: Multilabel

Domanda **3**

Completo

Punteggio ottenuto 2,00 su 2,00

Provide a detailed description of an LSTM cell.

LSTM (Long Short Term Memory) is a memory that was introduced to limit the vanishing gradient problem that happens in normal recurrent linear network, in particular when the sequence to analyse is very long. At each step, it allows to forget part of the memory during the training and allows new information to be stored. It is composed of three gates:

- the forget gate aims to forget part of the previous cell state to retain just the information needed. To do so, it computes a gate value through a sigmoid function and does an element-wise product with the previous cell state.
- the input gate computes both a gate value through a sigmoid function and a candidate value through a tanh function to obtain what we want to add to memory. The two values are in fact concatenated (element-wise product).
- the output gate updates the hidden state. After adding to the cell state obtained from the forget gate (with the information to keep), the information to add obtained from the input gate, it computes the updated hidden state. It uses a tanh function (that is applied to what I have just described) and then it does an element-wise product with a new gate value computed by the output gate itself (with a sigmoid function). The result is, in fact, the new hidden state.

Commento:

Domanda 4

Parzialmente corretta

Punteggio ottenuto 1,65 su 3,00

Select all and only those statements that are TRUE about the self-attention mechanism used in Transformer architectures.

- ☒ a. In self-attention, the matrices for queries, keys, and values are learned during training. ✓
- ☐ b. Self-attention is a core mechanism in natural language processing, but not applicable to computer vision tasks.
- ☐ c. Self-attention requires sequential token processing and therefore cannot be parallelized.
- ☒ d. Self-attention enables each token in the input sequence to attend to all other tokens, including itself. ✓
- ☒ e. Multi-head attention allows the model to focus on different aspects/perspectives of the input sequence simultaneously. ✓
- ☐ f. The softmax normalization ensures that attention weights are positive and sum to 1.
- ☒ g. The division by $\sqrt{d_k}$ in the attention formula prevents large dot products from dominating the softmax. ✓
- ☐ h. The attention weights are computed using the dot product of the value and query matrices.
- ☒ i. Self-attention layers are invariant to the order of input tokens, i.e., no matter the order of the input tokens the output is always the same sequence of tokens. Because of this Positional Encoding has been added. ✗

Your answer is partially correct.

Hai selezionato correttamente 4.

Le risposte corrette sono: Self-attention enables each token in the input sequence to attend to all other tokens, including itself., The softmax normalization ensures that attention weights are positive and sum to 1., In self-attention, the matrices for queries, keys, and values are learned during training., The division by $\sqrt{d_k}$ in the attention formula prevents large dot products from dominating the softmax., Multi-head attention allows the model to focus on different aspects/perspectives of the input sequence simultaneously.

Domanda 5

Parzialmente corretta

Punteggio ottenuto 0,50 su 1,00



Autotune is a pitch correction software used by singers to enhance their vocal performance in **real-time** while they are singing. It works by analyzing a singer's voice and adjusting it to match the correct notes. It can be used to create a variety of vocal effects, from subtle adjustments to more noticeable and dramatic results.

We want to use a neural network to implement it. Select which of the following architectures can be used to implement it (wrong answers have negative scores, the average score is 0).

- | | |
|---|---|
| ☒ a. A Bidirectional LSTM that takes as input streams the singer's voice and the target instrument, i.e., the one to adjust the pitch accordingly. | ☐ A Bidirectional LSTM is a batch model that can processes sequences not streams. |
| ☒ b. A Recurrent Neural Network that takes as input streams the singer's voice and the target instrument, i.e., the one to adjust the pitch accordingly. | ☒ It can be used, as for the vanishing gradient is a limited issue as the memory required to the system is rather limited |
| ☐ c. Assuming we can compute a Fast Fourier Transform fast enough, we can use a Feed Forward Neural Network which takes as input the spectrum of the singer's voice, and provides as output the target one | |

Your answer is partially correct.

Hai selezionato troppe opzioni.

La risposta corretta è: A Recurrent Neural Network that takes as input streams the singer's voice and the target instrument, i.e., the one to adjust the pitch accordingly.

Domanda 6

Parzialmente corretta

Punteggio ottenuto 0,67 su 1,00

Mark all the answers that are true about convolutional neural networks

. When answering a question:

- remember that our goal is to assess your understanding of the course materials, rather than your programming skills in Python. Our questions are not about what is "technically possible in Python", but rather about the fundamentals of the deep learning models and what are the best practices.
- act as a Neural Network expert, as you are expected to be, not as a Python programming expert.

Wrong answers result in negative points.

Scegli una o più alternative:

- ☐ a. Despite ReLU is a linear function, it is commonly used in many CNN architectures, as it improves convergence of gradient descent.
- ☐ b. There is no need to specify the number of channels of filter in a convolutional layer as these typically correspond to the number of channels of the input image.
- ☐ c. During training, the feature extraction component of a CNN optimizes also the pooling steps.
- ☐ d. The learnable component of the convolutional layer are only the filter weights and biases.
- ☐ e. Early CNN architectures had the majority of trainable parameters in the convolutional layers.
- ☐ f. FALSE: There is a specific reason for concatenating conv layers and MLP as feature classifier: we need to perform Back-propagation. Other classifiers can't be used to train the convolutional backbone of the CNN.
- ☐ g. There is no specific reason to use an MLP as a feature classifier after the flattening and the convolutional block of a CNN. Any classifier can be used in place of a MLP and the entire network can be trained end-to-end.
- ☒ h. The same convolutional layer, placed at different depth of the same CNN might result in a different number of parameters. ✔ TRUE, as the number of channels depends on the input activation to the conv layer.
- ☒ i. A convolutional layer can be seen as a linear classifier where the matrix has sparse rows. The nonzero entries among the rows are often repeated and shifted. ✔ TRUE, this is a direct consequence of weight sharing and sparse connectivity of CNNs

Your answer is partially correct.

Hai selezionato correttamente 2.

Le risposte corrette sono: The learnable component of the convolutional layer are only the filter weights and biases., A convolutional layer can be seen as a linear classifier where the matrix has sparse rows. The nonzero entries among the rows are often repeated and shifted., The same convolutional layer, placed at different depth of the same CNN might result in a different number of parameters.

Domanda 7

Parzialmente corretta

Punteggio ottenuto 1,67 su 2,00

Mark all the answers that are true about Data Augmentation

. When answering a question:

- remember that our goal is to assess your understanding of the course materials, rather than your programming skills in Python. Our questions are not about what is "technically possible in Python", but rather about the fundamentals of the deep learning models and what are the best practices.
- act as a Neural Network expert, as you are expected to be, not as a Python programming expert.

Wrong answers result in negative points.

Scegli una o più alternative:

- ☒ a. A criteria to consider when designing augmentation procedures is that the transformation should preserve the label of the input image. ✔ TRUE
- ☒ b. Data augmentation is a good way to train a neural network when there are limited training data. ✔ TRUE!
- ☐ c. Test time augmentation is a way to ease network convergence and increase robustness to mistakenly annotated samples.
- ☐ d. Test time augmentation adds no computational overhead, as the augmented images can be stacked along the channel dimension.
- ☐ e. In mixup augmentation, when the samples to be mixed belong to the same class, it is not possible to define the target to be predicted.
- ☐ f. Mixup augmentation cannot be used in classification problem, as the output associated to an augmented sample is not a discrete value, but rather the coefficients of the input mixture.
- ☐ g. It is possible to adopt label-specific transformation augmentation, both at training and testing time.
- ☒ h. When training a standard CNN on binary images, it is not possible to use augmentations like rotation as these might give rise to non binary entries. ✘ FALSE: this is not a problem for the network to have non binary images, as these are always casted in floating inputs in standard training settings
- ☒ i. It is possible to leverage data augmentation also at test time to improve invariance w.r.t. augmentation transformation. ✔ TRUE

Your answer is partially correct.

Hai selezionato troppe opzioni.

Le risposte corrette sono: Data augmentation is a good way to train a neural network when there are limited training data., A criteria to consider when designing augmentation procedures is that the transformation should preserve the label of the input image., It is possible to leverage data augmentation also at test time to improve invariance w.r.t. augmentation transformation.

Domanda 8

Parzialmente corretta

Punteggio ottenuto 0,67 su 1,00

Please answer the following questions about the Inception Block

Scegli una o più alternative:

- ☐ a. The inception block promotes sparsity in the feature maps.
- ☐ b. A network architecture having inception blocks does not need maxpooling layers, simple strided convolutions are sufficient thanks to this layer.
- ☒ c. The inception block leverages 1x1 convolutions to keep controlled the overall number of operations. ✓ TRUE
- ☐ d. In a variant of the inception block, 1x1 convolutional layers were introduced to reduce the overall number of parameters.
- ☐ e. The inception block reduces input dimension very efficiently
- ☐ f. The inception block does not increase the receptive field thanks to skip connections.
- ☒ g. The inception block promotes learning features at multiple scales from the input image. ✓ TRUE
- ☐ h. The inception block gives rise to activations maps of different spatial dimensions.

Your answer is partially correct.

Hai selezionato correttamente 2.

Le risposte corrette sono: The inception block promotes learning features at multiple scales from the input image., The inception block leverages 1x1 convolutions to keep controlled the overall number of operations., In a variant of the inception block, 1x1 convolutional layers were introduced to reduce the overall number of parameters.

Domanda 9

Risposta non data

Non valutata

Consider the following network architecture

```
import tensorflow as tf
```

```
tfkl = tf.keras.layers
```

```
tfk = tf.keras
```

```
def get_model_5(input_shape):
```

```
    input_layer = tfkl.Input(shape=input_shape, name='input_layer')
```

```
    x = tfkl.Conv2D(filters=32, kernel_size=3, padding='same', activation='relu', name='conv1')(input_layer)
```

```
    x = tfkl.BatchNormalization(name='bn1')(x)
```

```
    x = tfkl.MaxPooling2D(pool_size=2, strides=2, name='pool1')(x)
```

```
    x = tfkl.Conv2D(filters=64, kernel_size=3, padding='same', activation='relu', name='conv2')(x)
```

```
    x = tfkl.BatchNormalization(name='bn2')(x)
```

```
    x = tfkl.MaxPooling2D(pool_size=2, strides=2, name='pool2')(x)
```

```
    x = tfkl.Conv2D(filters=128, kernel_size=1, padding='same', activation='relu', name='conv3')(x)
```

```
    x = tfkl.BatchNormalization(name='bn3')(x)
```

```
    x = tfkl.Conv2D(filters=64, kernel_size=3, padding='same', activation='relu', name='conv4')(x)
```

```
    x = tfkl.UpSampling2D(size=2, name='up1')(x)
```

```
    x = tfkl.Conv2D(filters=32, kernel_size=3, padding='same', activation='relu', name='conv5')(x)
```

```
    x = tfkl.UpSampling2D(size=2, name='up2')(x)
```

```
    output_layer = tfkl.Conv2D(filters=input_shape[-1], kernel_size=3, padding='same', activation='linear',
name='output_layer')(x)
```

```
    model = tf.keras.Model(inputs=input_layer, outputs=output_layer)
```

```
    model.compile(
```

```
        optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
```

```
        loss=tf.keras.losses.MeanSquaredError(),
```

```
        metrics=[
```

```
            tf.keras.metrics.MeanAbsoluteError(name='mae'),
```

```
            tf.keras.metrics.RootMeanSquaredError(name='rmse')
```

```
        ]
```

```
    )
```

```
    return model
```

```
input_shape = (256,256,3)
```

```
model = get_model_5(input_shape=input_shape)
```

```
model.summary()
```

```
tfk.utils.plot_model(model, show_shapes=True, dpi=50)
```

The last command returns the following summary, which we provide for reference.

Feel free to copy and paste this command in the textbox below to prepare your answer, otherwise you can use plain paper.

Consider that this question won't be corrected, numbers need to be reported in the next one according to the **strict** guidelines provided

Layer (type)	Output Shape	Param #
input_layer (InputLayer)		
conv1 (Conv2D)		
bn1 (BatchNormalization)		
pool1 (MaxPooling2D)		
conv2 (Conv2D)		
bn2 (BatchNormalization)		
pool2 (MaxPooling2D)		
conv3 (Conv2D)		
bn3 (BatchNormalization)		
conv4 (Conv2D)		
up1 (UpSampling2D)		
conv5 (Conv2D)		
up2 (UpSampling2D)		
output_layer (Conv2D)		
Total params:		
Trainable params:		
Non-trainable params:		

Domanda 10

Risposta corretta

Punteggio ottenuto 4,00 su 4,00

Remember that this question will be automatically rated, therefore you have to consider the following details very carefully:

- when the activations become a vector, these are stacked along the third dimension (the channel, C). Therefore, if your latent representation becomes a 3 element vector, enter as sizes H:1 W:1 C:3.
- Enter digits **only** and do not use "." or "," or "-" or "[]". Say write "1234" and not "1,234" or "1.234". Write "0", not "-" or "[]".
- Do not enter formulas, just resulting numbers
- Do not enter empty spaces... please double check
- You don't need to fill in connections, but please consider them.
- Wrong answers result in negative points

All the numbers have to be entered here, the text in the previous question is just meant for you to design the solution.

Layer (type)	Output Shape	Param #	Connected to (omitted, not to be entered)
input_layer (InputLayer)	H: 256 ✓ , W: 256 ✓ , C: 3 ✓	Params #: 0 ✓	
conv1 (Conv2D)	H: 256 ✓ , W: 256 ✓ , C: 32 ✓	Params #: 896 ✓	
bn1 (BatchNormalization)	H: 256 ✓ , W: 256 ✓ , C: 32 ✓	Params #: 128 ✓	
pool1 (MaxPooling2D)	H: 128 ✓ , W: 128 ✓ , C: 32 ✓	Params #: 0 ✓	
conv2 (Conv2D)	H: 128 ✓ , W: 128 ✓ , C: 64 ✓	Params #: 18496 ✓	
bn2 (BatchNormalization)	H: 128 ✓ , W: 128 ✓ , C: 64 ✓	Params #: 256 ✓	
pool2 (MaxPooling2D)	H: 64 ✓ , W: 64 ✓ , C: 64 ✓	Params #: 0 ✓	
conv3 (Conv2D)	H: 64 ✓ , W: 64 ✓ , C: 128 ✓	Params #: 8320 ✓	
bn3 (BatchNormalization)	H: 64 ✓ , W: 64 ✓ , C: 128 ✓	Params #: 512 ✓	
conv4 (Conv2D)	H: 64 ✓ , W: 64 ✓ , C: 64 ✓	Params #: 73792 ✓	
up1 (UpSampling2D)	H: 128 ✓ , W: 128 ✓ , C: 64 ✓	Params #: 0 ✓	
conv5 (Conv2D)	H: 128 ✓ , W: 128 ✓ , C: 32 ✓	Params #: 18464 ✓	
up2 (UpSampling2D)	H: 256 ✓ , W: 256 ✓ , C: 32 ✓	Params #: 0 ✓	
output_layer (Conv2D)	H: 256 ✓ , W: 256 ✓ , C: 3 ✓	Params #: 867 ✓	
=====			
Total params:	121731 ✓		
Trainable params:	121283 ✓		
Non-trainable params:	448 ✓		

Model: "functional_3"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 256, 256, 3)	0
conv1 (Conv2D)	(None, 256, 256, 32)	896
bn1 (BatchNormalization)	(None, 256, 256, 32)	128
pool1 (MaxPooling2D)	(None, 128, 128, 32)	0
conv2 (Conv2D)	(None, 128, 128, 64)	18,496
bn2 (BatchNormalization)	(None, 128, 128, 64)	256
pool2 (MaxPooling2D)	(None, 64, 64, 64)	0
conv3 (Conv2D)	(None, 64, 64, 128)	8,320
bn3 (BatchNormalization)	(None, 64, 64, 128)	512
conv4 (Conv2D)	(None, 64, 64, 64)	73,792
up1 (UpSampling2D)	(None, 128, 128, 64)	0
conv5 (Conv2D)	(None, 128, 128, 32)	18,464
up2 (UpSampling2D)	(None, 256, 256, 32)	0
output_layer (Conv2D)	(None, 256, 256, 3)	867

Total params: 121,731 (475.51 KB)

Trainable params: 121,283 (473.76 KB)

Non-trainable params: 448 (1.75 KB)

Domanda 11

Parzialmente corretta

Punteggio ottenuto 1,50 su 2,00

Mark all the answers that are true. When answering a question:

- assume that the network is provided with all the labeling information needed for the described training
- consider that our goal is to assess your understanding of the course materials, rather than your programming skills in Python. Our questions are not about what is "technically possible in Python", but rather about the fundamentals of the deep learning models.
- act as a Neural Network expert, as you are expected to be, not as a Python programming expert.

Wrong answers result in negative points.

Scegli una o più alternative:

- ☐ a. Given a natural image picture, the network segments all the persons in there, covering with different colours each person.
- ☒ b. Given a picture affected by a random rotation, the network returns the correctly oriented image ✓ TRUE:
- ☐ c. Given an aerial picture where each pixel corresponds to 10mx10m. The network returns in each pixel the area covered by vegetation, the area covered by water and the area covered by roads.
- ☒ d. This network can be trained to suppress noise in aerial images, like those from Google Earth. ✓ TRUE:
- ☐ e. Given a picture of an animal, the network segments his eyes, nose and tail
- ☐ f. This is an object detection network that returns a list of bounding boxes and to draw colored bounding boxes around each object in the image.
- ☐ g. The network classifies each pixel of an histological image into three categories: cell, tissue no-cell, no tissue.
- ☐ h. Given a picture containing multiple geometric shapes, the network returns a list of all the vertices.
- ☐ i. This is a network that takes as input an RGB image and predicts whether it contains an object, an animal or a human
- ☐ j. The network predicts a probability map for an input image (e.g. the probability of having rainfalls in each pixel of a map)
- ☒ k. Given an RGB picture corrupted by blur, the network restores the unblurred picture, without providing any information on the blurring function. ✓ TRUE:

Your answer is partially correct.

Hai selezionato correttamente 3.

Le risposte corrette sono: Given a picture affected by a random rotation, the network returns the correctly oriented image, This network can be trained to suppress noise in aerial images, like those from Google Earth., Given an RGB picture corrupted by blur, the network restores the unblurred picture, without providing any information on the blurring function., Given an aerial picture where each pixel corresponds to 10mx10m. The network returns in each pixel the area covered by vegetation, the area covered by water and the area covered by roads.